

MARCOS RODRIGO TALAVERA



Ph.D. Telecommunication Engineer and AI Researcher specializing in deep learning for computer vision, with expertise in face recognition, video understanding, and vision-language models. Full profile, projects and publications at marcosrodrigot.github.io

CONTACT

- ✉ marcosrodrigo5@hotmail.com
- ☎ +34 618 382 472
- 📍 Madrid, Spain
- 🏠 marcosrodrigot.github.io
- 🐦 @MarcosRodrigoT
- 📞 0000-0002-1808-4738

SKILLS

AI & Deep Learning

PyTorch

TensorFlow

HuggingFace

OpenCV

LangChain

Ollama

ML & Data Science

Scikit-learn

NumPy / Pandas

Weights & Biases

Programming

Python

C / C++

Bash

SQL

HTML / CSS / JS

LaTeX

Infrastructure & Tools

Linux

Docker

Kubernetes

AWS

Git

Languages

Spanish (Native)

English (C1 Certified)

French (Basic)

SUPPLEMENTARY EDUCATION

- How Diffusion Models Work (2025), DeepLearning.AI
- Generative AI with LLMs (2024), DeepLearning.AI, Coursera, AWS
- Deep Learning Specialization (2023), DeepLearning.AI, Coursera
- English as a Second Language (2015–16), UCLA
- Campus Party Tech. Event (2013), UAH
- Cognitive Development (2010–12), SAPIENTEC
- Mathematical Thinking (2006–09), Esc. Miguel de Guzmán

EDUCATION

- 📅 2020 – 2025

📍 Universidad Politécnica de Madrid, UPM

Ph.D. in Communication Technologies

Thesis: Automatic Sports Video Summarization with Identity-Aware Highlight Selection.
- 📅 2018 – 2020

📍 Universidad Politécnica de Madrid, UPM

M.Sc. in Telecom. Engineering

Thesis: Automatic Highlight Detection in Martial Arts Tricking Videos. Grade: 10/10.
- 📅 2013 – 2018

📍 Universidad de Alcalá de Henares, UAH

B.Sc. in Telecom. Systems Eng.

Thesis: Acoustic Bird Classification Using MFCC Feature Extraction. Grade: 9/10.

PROFESSIONAL EXPERIENCE

- 📅 2020 – present

📍 E.T.S.I. Telecomunicación, UPM

Post-graduate Researcher

Designed deep-learning pipelines for face recognition (CNNs, ViTs), video highlight detection (2D/3D CNNs, RNNs, Transformers) and multimodal video summarization (CLIP). Led R&D with Nokia and Airbus. Taught across three degree programmes, supervised two Master's theses and built a 17-node Kubernetes GPU cluster.
- 📅 2018

📍 Urban Planet

Sports Instructor

Led group classes, ensured safety protocols and provided customer service in English.
- 📅 2017 – 2018

📍 Indra Sistemas S.A.

Junior Programmer

Developed VoIP scripts for air-traffic management. Automated tasks and configured network equipment under Scrum.
- 📅 2016 – 2017

📍 DEYDE Calidad de Datos S.L.

Data Coder & Trainee Programmer

Built expert systems for postal-address coding and maintained master databases.
- 📅 2013 – 2018

📍 Self-employed

Private Tutor

Taught Maths, Physics, Chemistry and English to students preparing for university entrance.

PUBLICATIONS

- M. Rodrigo**, E. Zhong, C. R. del-Blanco, C. Cuevas, F. Jaureguizar, N. García. “ViMoCLIP: Augmenting static CLIP representations with video-motion cues for animal-action recognition.” *IEEE/CVF CVPR Workshops (CV4Animals)*, 2025.
- M. Rodrigo**, C. Cuevas, N. García. “Text-guided sports highlights: a CLIP-based framework for automatic video summarization.” *IEEE Access*, 2025.
- M. Rodrigo**, C. Cuevas, N. García. “Comprehensive comparison between vision transformers and CNNs for face recognition.” *Sci. Reports* 14, 21392, 2024.
- M. Rodrigo**, C. Cuevas, D. Berjón, N. García. “Automatic highlight detection in videos of martial arts tricking.” *Multimed. Tools Appl.* 83, 17109–17133, 2024.
- M. Rodrigo**, E. González-Sosa, C. Cuevas, N. García. “UPM-GTI-Face: Impact of distance and masks in face detection and recognition.” *IEEE AVSS*, pp. 1–8, 2022.

AREAS OF EXPERTISE

Deep Learning

Computer Vision

Multimodal Learning

Face Recognition

Video Summarization

Action Recognition

Vision Transformers

CLIP / VLMs

Object Detection

Diffusion Models

LLMs

GPU Clusters

Teaching